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Globepay payments: acceptance, declines and chargebacks

Analytics Engineer take-home - dbt Core on BigQuery

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One tested dbt model answers all three questions

Q1: ACCEPTANCE RATE

69.6%

of payment attempts accepted, Jan–Jun 2019. Stable, no trend: monthly between 67.8% and 71.7%.

Q2: DECLINED OVER $25M

4 of 6

countries: FR, UK, AE and US. If amounts are already in USD: AE, CA, US (slide 6).

Q3: MISSING CHARGEBACK DATA

0

attempts without chargeback data. But recent chargebacks are provisional: the report has no date.



fct_globepay_transactions one row per payment attempt (5,430) - chargebacks joined - USD amounts

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PART 1

Ingestion and modeling

Exploration - architecture - lineage - validation - tips

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Raw data lands untouched; dbt does all the typing

01

2 CSV reports

Acceptance (11 columns) and chargeback (4 columns), 5,430 rows each. API spec used as data dictionary.

02

bq load, all STRING

No schema inference: nothing is parsed or rejected at load time.

03

raw_globepay

Two raw tables declared as dbt sources, with tests. Not seeds.

04

Typed in staging

Casts, renames and FX conversion are versioned, tested and documented.

Why STRING first: a malformed value can never break the load, and every cast is visible in code review and covered by a test.

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Eleven profiling queries shaped the model

Clean grain

external_ref is unique and never null in both reports: one row per payment attempt.

Full join coverage

All 5,430 attempts are in both reports. No orphan chargebacks.

IDs swapped vs the spec

The spec's evt_... external_ref is in the CSV ref column. The chargeback report joins on CSV external_ref (5,430/5,430), so that is the key.

One negative amount

-23.78 GBP on an accepted attempt: kept as reported, flagged by a warn test.

FX rates as JSON

Local units per 1 USD, one rate a day. amount_usd = amount / rates[currency].

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Open question: what unit and currency is amount in?

The API spec says

amount in minor units: `1000` = $10.00

The CSV shows

two decimals, e.g. `1020.46`, and the same distribution in every currency

Decision: major units, local currency

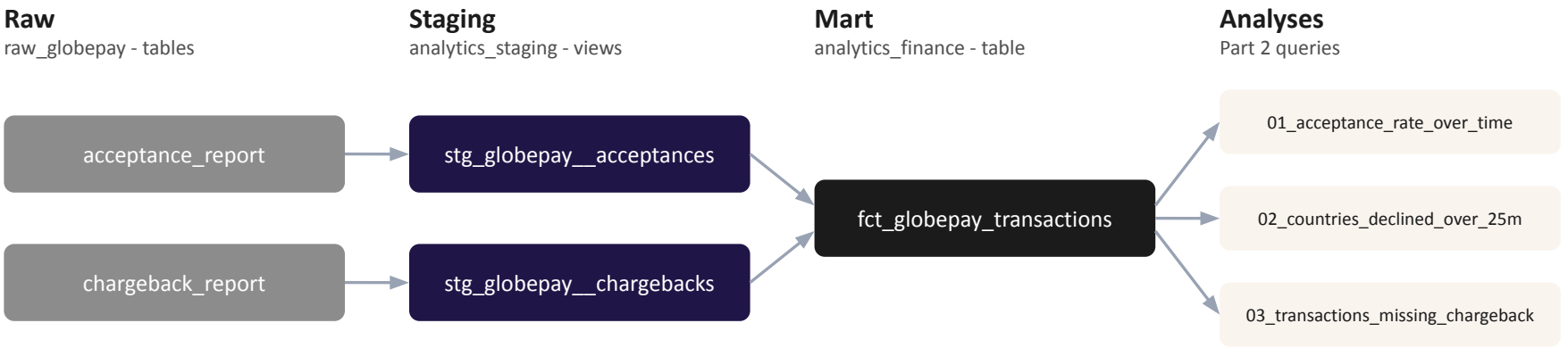
- Decimals rule out minor units (5,382 of 5,430 values)
- The spec pairs amount with its own currency; USD settlement is a separate step
- Hints only, not proof: the $25M threshold and the rates field

Three readings, three Q2 answers

- Local currency, converted (chosen): FR, UK, AE, US
- Already in USD: AE, CA, US
- Minor units: none (max $0.33M)
- Same distribution in every currency: confirm with Globepay

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Three layers: sources, staging, one mart



Staging

One model per source, 1:1 with its grain. Rename, cast, row-level logic only: no joins, no filters.

Mart

Acceptances LEFT JOIN chargebacks. A table in a finance dataset: the single model the analyst queries.

No intermediate layer

One join and no reusable logic yet. Add one (ephemeral) when a second processor arrives.

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Design choices that keep the mart honest

Consistent naming

*_id keys, *_at timestamps, is_/has_ booleans, *_code codes, amount_<currency>.

USD conversion in staging

Row-level logic with the rate Globepay applied, so every model shares one amount_usd.

LEFT JOIN + explicit flag

has_chargeback_data separates “no row” from FALSE; is_chargeback stays NULL when unknown.

Mart as a table

Fast and cheap for the analyst, rebuilt in seconds. Incremental is not needed yet.

Docs in the warehouse

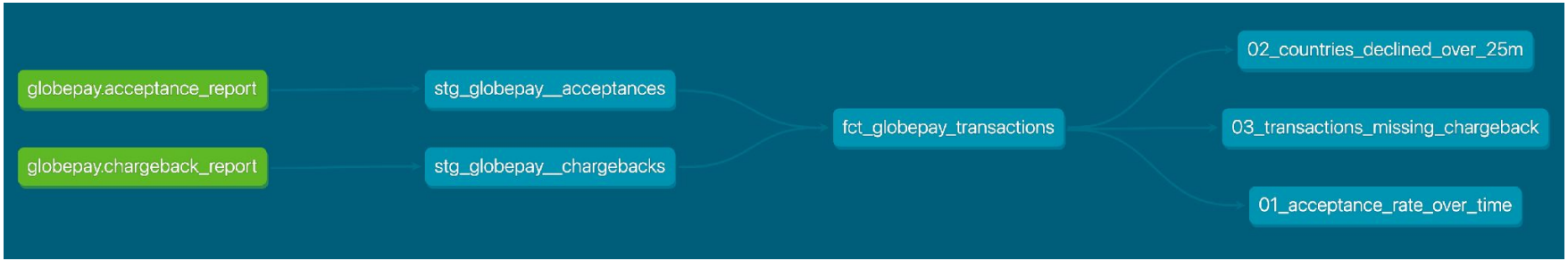
persist_docs writes descriptions to BigQuery, where the analyst actually queries.

Portable to Snowflake

Only four BigQuery-specific functions: JSON parsing, CAST to BOOL, COUNTIF, DATE_TRUNC.

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Every mart column traces back to a raw field



dbt docs lineage, selection +fct_globepay_transactions+

Mart column	Raw column	Transformation
`transaction_id`	`acceptance_report.external_ref`	rename
`is_accepted`	`state`	state = 'ACCEPTED'
`amount_usd`	`amount, rates, currency`	amount / rates[currency]
`has_chargeback_data`	`chargeback_report.external_ref`	IS NOT NULL after the LEFT JOIN
`is_chargeback`	`chargeback`	CAST AS BOOL

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30 data tests, one job per layer

30

data tests run on every dbt build

error when a failure makes the model wrong

warn for business anomalies (new country, negative amount, missing chargeback row)

Sources **Raw data matches what the models assume**

unique / not_null on external_ref; relationships both ways (acceptances → chargebacks at warn, for Q3); not_null + accepted_values on state and chargeback

Staging **Casts and derived columns are valid**

not_null on cast columns; accepted_values on country, currency, source (warn); FX rate > 0; amount_local ≥ 0 (warn)

Mart **The grain and the join are correct**

unique / not_null on transaction_id; equal_rowcount vs staging; is_chargeback not_null where has_chargeback_data

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Tips on macros and documentation

Macros

- generate_schema_name_for_env to split dev and prod datasets
- to_usd() so every future Globepay report converts the same way

```
{% macro to_usd(amount, currency, rates) -%}
    CAST({{ amount }} / LAX_FLOAT64(
        PARSE_JSON(
            {{ rates }}
            , wide_number_mode => 'round'
        )[{{ currency }}]
    ) AS NUMERIC)
{%- endmacro %}
```

Documentation

In this project

- Every model and column described in YAML, edge cases included (UK, negative amount, NULL = unknown)
- persist_docs: descriptions visible in BigQuery
- A guide for the analyst with caveats and example queries

Next

- Doc blocks for columns repeated across models
- Exposures for the dashboards that read the mart

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PART 2

Answers for the Data Analyst

Q1 acceptance rate - Q2 declined over $25M - Q3 missing chargeback data

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fct_globepay_transactions: one row per attempt

Column	Use it for
`transaction_id`	Primary key
`transaction_date`	Time grain: DATE_TRUNC(…, ISOWEEK)
`country_code`	Breakdowns (UK, not GB)
`is_accepted`	Acceptance rate
`amount_usd`	Comparing and summing across countries
`has_chargeback_data`	FALSE = no row in the chargeback report
`is_chargeback`	TRUE / FALSE / NULL (see right)

Reading is_chargeback

TRUE Charged back

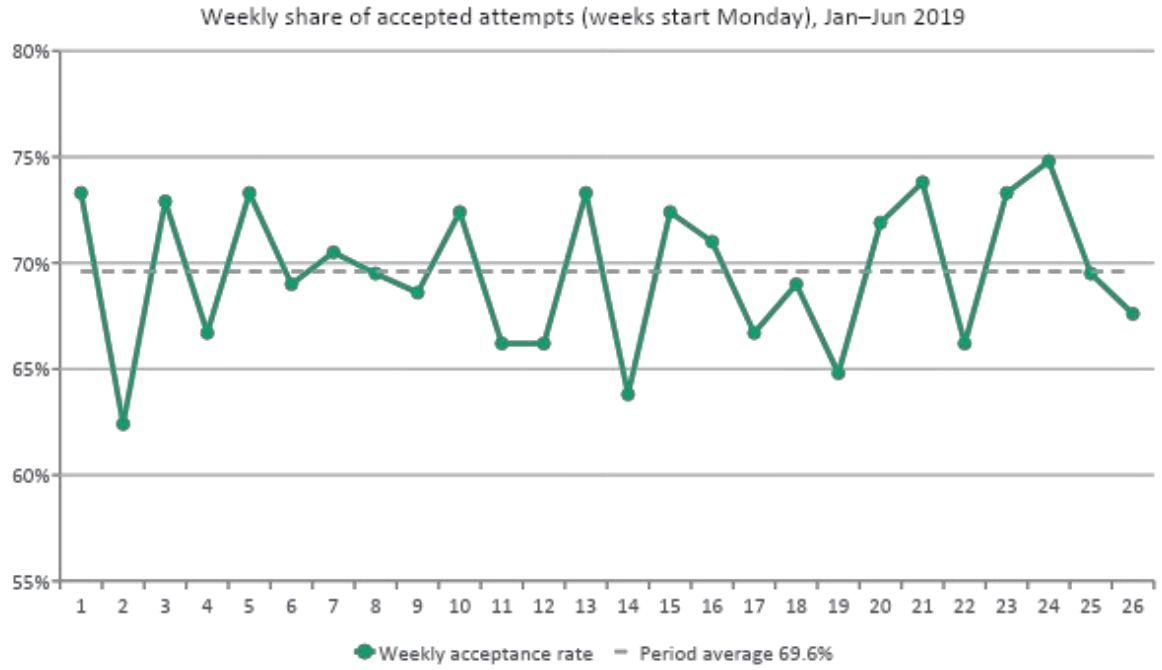
FALSE Not charged back (not applicable if declined)

NULL Unknown: no chargeback data. Filter on has_chargeback_data first

Always use amount_usd across countries: amount_local only compares within one currency.

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Q1 - Acceptance holds at about 70% with no trend



69.6%

3,777 of 5,430 attempts accepted

- Monthly: 67.8% to 71.7%
- Weekly: 62.4% (7 Jan) to 74.8% (10 Jun)
- About 210 attempts a week: ±3 points is one standard deviation. Noise, not a change in behaviour

First week has 6 days (from 1 Jan). Source: fct_globepay_transactions.

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Q1 - No country stands out: a 4-point spread is noise

Country	Jan	Feb	Mar	Apr	May	Jun	Jan–Jun
UK	69.0%	69.3%	67.7%	72.0%	74.2%	76.7%	**71.5%**
MX	74.2%	67.9%	69.7%	64.7%	74.2%	76.0%	**71.2%**
FR	67.7%	75.7%	74.2%	67.3%	67.1%	68.7%	**70.1%**
CA	74.8%	74.3%	67.7%	66.7%	66.5%	68.0%	**69.6%**
AE	65.2%	69.3%	69.7%	64.7%	67.7%	70.7%	**67.8%**
US	66.5%	64.3%	64.5%	71.3%	66.5%	70.0%	**67.2%**
All	**69.6%**	**70.1%**	**68.9%**	**67.8%**	**69.4%**	**71.7%**	**69.6%**

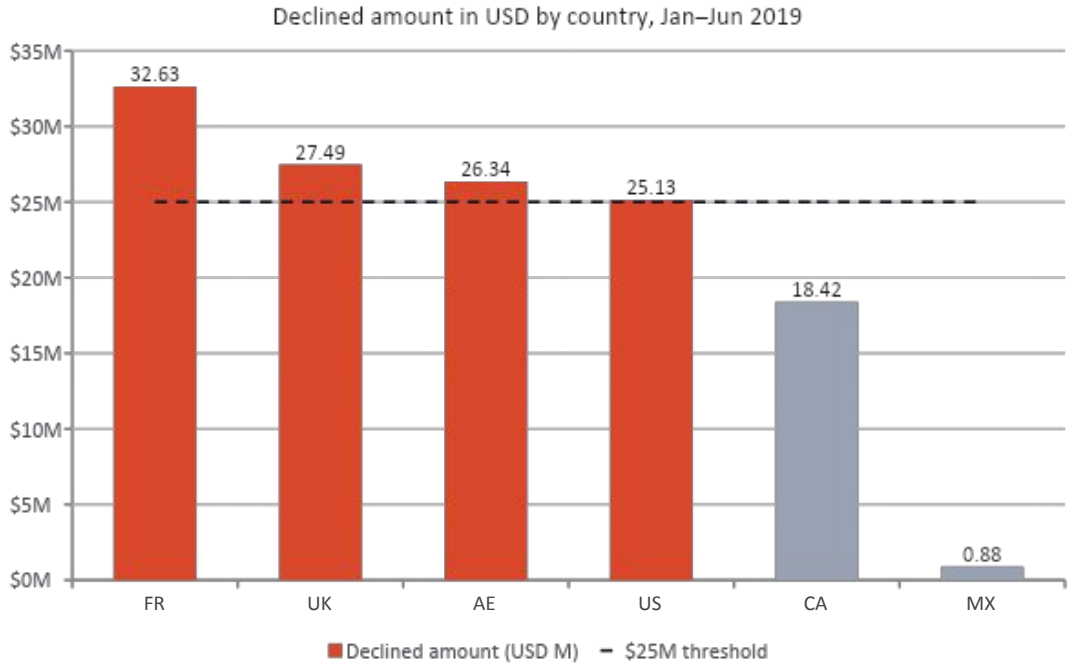
4 pts

US 67.2% to UK 71.5%, Jan–Jun

- Chi-squared across the six countries is 6.5, below the 11.1 needed at 5%: within chance
- One country-month is about 150 attempts: ±3.7 points is one standard deviation
- Worth watching, not a trend yet: UK 68.7% → 74.3% and CA 72.2% → 67.0% (Jan–Mar vs Apr–Jun), neither significant

Monthly acceptance rate by card country, 905 attempts per country. Source: 01_acceptance_rate_by_country.sql.

Q2 - Four of six countries passed $25M declined



Converted with the rate Globepay applied on each day. Source: fct_globepay_transactions.

FR	**$32.6M**	271 declined
UK	**$27.5M**	258 declined
AE	**$26.3M**	291 declined
US	**$25.1M**	297 declined

Below: CA $18.4M and MX $0.9M.

The US is $0.13M over. In USD, so FX does not move it.

FX decides the UK. EUR and GBP fall about 25% vs USD; at 1 Jan rates the UK totals $24.0M.

Unit risk. If amount is already in USD, the answer is AE, CA, US (slide 6).

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Q2 - All four countries crossed $25M in the last 18 days

Country	Declined USD, Jan–Jun	Went over $25M on	Days before 30 Jun
FR	32.63M	13 Jun	17
UK	27.49M	22 Jun	8
AE	26.34M	27 Jun	3
US	25.13M	30 Jun	0
CA	18.42M	not reached	–
MX	0.88M	not reached	–

18 days

13–30 Jun: all four crossings

- The US crossed with its last declined attempt (30 Jun, 09:36)
- At the end of May no country was over $25M
- $25M is a total over six months, so the list depends on the period. For a recurring check, use a monthly or quarterly threshold

Running total of declined amount_usd per country, in time order. Source: 02_date_declined_crossed_25m.sql.

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Q3 - No transaction is missing chargeback data

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of 5,430 payment attempts have no row in the chargeback report

FALSE on a declined attempt means “not applicable”

A declined payment is never charged, so it cannot be charged back. Compute chargeback rates on accepted attempts only: 223 of 3,777 = 5.9%.

Having data is not the same as final

Chargebacks arrive weeks after the payment and the report has no date. Treat the latest weeks as provisional; a load timestamp and freshness checks would make this visible.

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Next steps

Confirm with Globepay

Unit and currency of amount (changes Q2), which ID is external_ref, AE charged in USD.

Unit tests

Fixed rows (USD, EUR, a negative amount) to lock the direction of the FX conversion.

Freshness for chargebacks

_loaded_at on ingest and source freshness, so late reports alert and recent rates are flagged.

Dev/prod and CI

Separate datasets per environment; dbt build --select state:modified+ on every PR.

Doc blocks and exposures

Write shared column docs once; register the dashboards that read the mart.

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Thank you

Questions and discussion